iv) The total value of the resistance is reduced by adding one more resistor in the circuit. If resistances are connected in parallel then the total resistance

$$R = \frac{R_1R_2R_3}{R_2R_3+R_1R_3+R_1R_2}$$

v) If there is a fault in any one resistor, the other two resistors will work because the current will flow through these resistors.

Example: In our home or factories, different types of electrical appliances are used, i.e, fan, bulb, television, motor, heater, etc. Each one has a separate circuit. For example if there is a fault in fan circuit, the fan circuit alone is inactive. Other circuits like bulb, television etc will work continuously. So, in parallel circuits there are a number of current paths available.

PROBLEMS – (Series Circuit)

1) ***10Ω, 20Ω and 30Ω resistances are connected in series. The circuit voltage is 240V. Calculate the i) Total resistance ii) current of this circuit.***

$R_1 = 10\Omega, R_2 = 20\Omega, R_3 = 30\Omega$
$V = 240V$
$R = ?$
$I = ?$

When the resistors are connected in series
Then $R = R_1 + R_2 + R_3$
$=10 + 20 + 30 = 60\Omega$

Total Resistance $R = 60\Omega$
According to ohm's law

$$I = \frac{V}{R}$$
$$= \frac{240}{60} = 4A$$

Current I = 4A.

2) ***Three resistances 5Ω, 15Ω and R_3 are connected in series. Resistance of this circuit is 60Ω. Find the value of R_3.***

$R = 60\Omega, R_1 = 5\Omega, R_2 = 15\Omega, R_3 = ?$

When the resistance are connected in series

Then $R = R_1 + R_2 + R_3$

$60 = 5 + 15 + R_3$ and
$60 = 20 + R_3$ $R_3 = 60-20 = 40\Omega$.

PROBLEMS – (Parallel Circuits)

1) ***Two resistances 8Ω and 2Ω are connected in parallel. Voltage of this circuit is 240V. Find the value of***
i) Total Resistance
ii) Current.

$R_1 = 8\Omega, R_2 = \Omega$
$R = ?$

In parallel circuit

$$R = \frac{R_1R_2}{R_1+R_2}$$
$$= \frac{8\times 2}{8+2} = \frac{16}{10} = 1.6\Omega$$

According to ohm's law

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